

Association between polychlorinated biphenyls and Parkinson's disease neuropathology

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Polychlorinated biphenyls (PCBs) are synthetic chemicals primarily used as coolants and insulators in electrical equipment. Although banned for several decades, PCBs continue to exist in the environment because of their long half-life, continued presence in items produced before the ban, and poor disposal practices. Epidemiological and experimental studies have identified exposure to PCBs as a potential risk factor for Parkinson's disease, perhaps more so in females. The objective of this work was to examine the association between PCB levels in post-mortem human brain tissue and the diagnosis of Parkinson's disease, as well as the degree of nigral depigmentation. We also sought to determine if this association was more significant when patients were stratified by sex. Post-mortem brain samples from control patients and those diagnosed with Parkinson's disease were obtained from the Emory University Brain Bank and from the Nun Study. Concentrations of eight prevalent PCB congeners were extracted from post-mortem brain tissue and analyzed using gas chromatography-mass spectrometry. PCB congeners 153 and 180 were significantly elevated in the brains of Parkinson's disease patients. When stratified by sex, the female Parkinson's disease group demonstrated significantly elevated concentrations of total PCBs and specifically congeners 138, 153, and 180 compared to controls, whereas PCB concentrations in males were not significantly different between control and Parkinson's disease groups. In a separate population of women (Nun Study) who had no clinical signs or symptoms of PD, elevated concentrations total PCB and congeners 138, 153 and 180 were also observed in post-mortem brain tissue exhibiting moderate nigral depigmentation compared to subjects with mild or no depigmentation. These quantitative data demonstrate an association between brain PCB levels and Parkinson's disease-related pathology. Furthermore, these data support epidemiological and laboratory studies reporting a link between PCB exposure and an increased risk for Parkinson's disease, including greater susceptibility of females.